南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Advanced Programming

Lab 1, **C/C++** environment configuration

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Topics

- Install C and C++ compilers
  - Windows: WSL, and then install gcc/g++ in WSL
  - macOS: clang
  - Linux: gcc/g++ or clang
- Verify your compilers
  - ➢ Example

```
➜  ~ gcc --version
Apple clang version 16.0.0 (clang-1600.0.26.6)
Target: arm64-apple-darwin25.3.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
➜  ~ g++ --version
Apple clang version 16.0.0 (clang-1600.0.26.6)
Target: arm64-apple-darwin25.3.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
```

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Exercises 1

Write a C program that initializes three double-precision floating-point variables with the values 0.1, 0.2, and 0.3, respectively. Then, use the `fprintf()` function to print each variable to standard output, formatting them to display exactly two decimal places.

```
→  examples git:(main) ✗ gcc main.c -o main && ./main
0.10
0.20
0.30
```

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Exercises 2

Put the provided code into three files, then compile and link them to produce an executable. Identify and fix any bugs that may be present in the code.

Steps:

1. Compile main.cpp into an object file.
2. Compile add.cpp into an object file.
3. Link the two object files to generate the final executable.

main.cpp

```
#include <iostream>

#include "Add.h"

int main()

{

    int num1 =    2147483647;

    int num2 =    1;

    int result =    0;

    result = add(num1, num2);

    cout << “The result is ” << result << endl;

    return 0;

}
```

add.h

```
#pragma once

int add(int n1, int n2);
```

add.cpp

```
#include "add.h"

int Add(int number1, int number2);

{

return n1 + n2;

}
```

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